

Congenital Heart Diseases (CHD)

Overview · etiology · fetal circulation · presentations · L→R shunts (ASD, VSD, PDA) · R→L shunts (TOF, TGA) · duct-dependent neonate

Yellow = stressed by the doctor · **Blue** = other high-yield · **Red underline** = clue that points to the diagnosis

1. Overview & Etiology

1.1 Recent Developments

- Lesions ↑ detected on **antenatal US**
- Echocardiography** = mainstay of diagnosis
- MRI** → 3D view of complex lesions, hemodynamics, flow → ↓ need for catheterization
- Most defects corrected at the **initial operation** (used to need 3–4 surgeries; now one surgery, catheter, or even meds)
- 60%** treated non-invasively (e.g. PDA)
- New therapies for **pulmonary hypertension** → delay transplant (it used to need lung transplant; now avoidable)
- Infant cardiac surgical mortality: **20% (1970) → 1.8% (2016)**

1.2 Epidemiology

- 8 / 1000** live births → significant CHD; **30%** need intervention in 1st year
- Some CVS abnormality (e.g. **bicuspid aortic valve**, can be normal) in **1–2%**
- 1 in 10 stillborn** infants have a cardiac anomaly (undetected CHD)
- 10–15% complex** (> 1 cardiac lesion; simple = one lesion)
- 10–15% also have a **non-cardiac** abnormality → always check other systems (genitalia, limbs, lungs, liver, brain)
- VACTERL association**: Vertebra · Anus · Cardiac · Tracheal fistula · Esophagus · Renal

1.3 Most Common Lesions: 5 Groups

Group	Lesions	Meaning
1) L→R shunt (breathless)	VSD 30% · PDA 12% · ASD 7%	<u>SOB, no cyanosis</u> ; systemic → pulmonary
2) R→L shunt (blue)	TOF 5% · TGA 5%	<u>Cyanosis</u> ; pulmonary → systemic
3) Common mixing (breathless + blue)	Complete AVSD 2%	Both
4) Outflow obstruction (vessels leaving the heart), well child	Pulmonary stenosis 7% · aortic stenosis 5%	Asymptomatic + murmur
5) Outflow obstruction, sick neonate	Coarctation of aorta 5%	<u>Collapse, cardiogenic shock</u>

1.4 Etiology (no single cause; risk factors)

Cause	Cardiac lesion	%
Maternal disorders		
Rubella	Peripheral pulmonary stenosis, PDA	30–35
SLE	<u>Complete heart block</u> (anti-Ro, anti-La)	35
Diabetes (uncontrolled in pregnancy)	↑ overall incidence	2
Maternal drugs (teratogenic)		
Warfarin	Pulmonary valve stenosis, PDA	5
Fetal alcohol syndrome	ASD, VSD, TOF	25
Chromosomal (all trisomies come with CHD)		
Down (T21)	AVSD , VSD	30
Edwards (T18) / Patau (T13)	Complex (> 1 lesion)	60–80 / 70
Turner (45XO)	Aortic valve stenosis, coarctation	15
22q11.2 deletion	Aortic arch anomalies, TOF, common arterial trunk	80
Williams (7q11.23)	<u>Supravalvular aortic stenosis</u> , peripheral PA stenosis	85
Noonan (PTPN11)	HCM, ASD, pulmonary valve stenosis	50

2. Fetal Circulation, Normal Heart & Presentations

2.1 Fetal Circulation

Placenta (oxygenation) → **umbilical vein** (oxygenated; outside the baby)
→ **ductus venosus** (inside the baby) → joins IVC → RA
→ **foramen ovale** (one-way, R→L) → LA → LV → aorta
▶ SVC blood → RV → PA → **ductus arteriosus** (RV → aorta) → bypasses lungs (high pulmonary pressure, amniotic fluid)
→ Body → **umbilical arteries** (deoxygenated) → placenta

- Main differences from adult: **oxygenation site** · **foramen ovale** · **ductus arteriosus**

2.2 Normal CXR & Hemodynamics

- Black = air (lungs, stomach); white = blood, bone (heart)
- Anterior border = mainly **RV**; posterior = mostly left heart
- RV enlarged → pushes LV posteriorly; LV normally not seen → enlarged → left heart border bulges **down and out**
- Pressure higher on the **left** → any opening → flow **L → R**

	RA	RV	PA	LA	LV	Aorta
Pressure	3	25/3	25/10	8	100/8	100/60
Sat %	75	75	75	98	98	98

2.3 5 Presentations of CHD

1. Antenatal US (before birth)
2. **Heart murmur** → **most common presentation**
3. Heart failure → can be the 1st presentation, with no murmur
4. Shock (cardiogenic)
5. Cyanosis → depends on hemodynamics

2.4 Cyanosis

- **Peripheral** (hands, feet) → cold, unwell, polycythemia; seen in acyanotic lesions (VSD, ASD)
- **Central** → **slate-blue tongue** → true cyanosis (TOF, TGA)
- Pulse oximeter: normal sat **≥ 94%**

2.5 Heart Murmur

- Not all murmurs = CHD; not all CHD has a murmur
- **Innocent (physiological)** murmur in **30%** of children
- Innocent = **4 S's** ("innoSent"): **a**symptomatic · **S**oft blowing (low grade) · **S**ystolic only · left **S**ternal edge
- **Diastolic murmur = always pathological**
- Systolic → heard with the pulse; diastolic → between pulses
- Innocent = **diagnosis of exclusion** → ECG, echo, CXR: normal + asymptomatic → normal murmur

2.6 Heart Failure (same as adults)

Symptoms

- Breathlessness, esp. on **feeding/exertion** (hard to judge: normal infant RR up to 60)
- **Sweating while feeding**
- **Poor feeding** → failure to thrive (normal feed ~20 min; in HF stops after 2–4 min)
- Recurrent chest infections
- **2 vitals (↑ RR, ↑ HR) + 2 organomegaly (heart, liver) = HF** (adults + peds)

Signs

- Poor weight gain / faltering growth
- **Tachypnea, tachycardia**
- Murmur, **gallop rhythm** (not a murmur)
- **Cardiomegaly, hepatomegaly**
- Cool peripheries

2.7 Causes of HF by Age

Age	Mechanism	Causes
Neonates (first 30 d)	Duct-dependent systemic circulation (outflow obstruction → collapsed baby, shock)	Hypoplastic left heart · critical aortic stenosis · severe coarctation (common) · interrupted aortic arch
Infants (2nd mo – 1 y)	High pulmonary blood flow	VSD (mostly) · AVSD · large PDA
Older (4–5 y +)	Right or left HF	Eisenmenger (right HF only) · rheumatic (common) · cardiomyopathy

3. Left-to-Right Shunts: Atrial Septal Defect (ASD)

3.1 Types

	Secundum ASD	Primum ASD (partial AVSD)
Frequency	80%, most common	Less common
Site	Central interatrial septum, away from valves	Lower septum, near AV junction → affects mitral + tricuspid valves
Management	Easier → catheter device	Open-heart surgery

- Both have similar symptoms and signs; anatomy and management differ
- Complete AVSD = different lesion (common mixing group)

3.2 Pathophysiology

↑ LA pressure → **L→R shunt** → ↑ venous return to RA/RV
→ **RV volume overload** → RV dilatation → ↑ pulmonary blood flow
→ **shortness of breath; no cyanosis**

3.3 Clinical Features

Symptoms

- **None** (commonly) → most kids with CHD are asymptomatic
- Recurrent chest infections / **wheeze** → cardiac wheeze (on echo), unlike asthma, no tachypnea
- **Arrhythmias** from the 4th decade
- Scenarios: wheezing child + abnormal echo → ASD · 40-y-old with arrhythmia + recurrent chest infections → ASD

Signs

- **Ejection systolic murmur, upper left sternal edge** → ↑ flow across pulmonary valve
- **Fixed, widely split S2** ("lub-dub-dub") → RV stroke volume equal in inspiration and expiration
- Split S2 = abnormal heart sound, **not a murmur** (S1 = mitral + tricuspid closure; S2 = aortic + pulmonary)
- Partial AVSD → also **apical pansystolic murmur** (AV valve regurgitation)

3.4 CXR (all L→R shunts)

- 1) **Cardiomegaly** 2) **congested lungs** (whitish network, ↑ pulmonary vascular markings) 3) **enlarged pulmonary arteries**
- CXR only tells L→R vs R→L, **not the exact defect**
- **Cardiomegaly = cardiothoracic ratio** > 60% in 1st year, > 50% after 1st year

3.5 Management

- Treat if **significant** (large enough to dilate the RV) → depends on size + clinical picture
- **Secundum → cardiac catheterization + occlusion device** (via femoral vein)
 - ▶ **Stable** (growing well, normal weight/height, no symptoms) → can wait to **3–4 y**
 - ▶ **Unstable** (affects growth, feeding, heart/lung enlargement) → must close even if small
- **Primum / partial AVSD → open-heart surgery** (affects the valves)
- Timing **3–5 y** → prevents right HF + arrhythmias later; no need to rush, let the baby grow

4. Left-to-Right Shunts: Ventricular Septal Defect (VSD)

4.1 Overview

- **30% of CHD = most common** (common in children)
- Defect anywhere in the ventricular septum
- Types: **perimembranous** (next to tricuspid valve; most common) · **muscular** (surrounded by muscle)
- Classified for management by **size vs the aortic valve**

4.2 Small vs Large VSD

	Small	Large
Size	< aortic valve ($\leq \sim 3$ mm); may be multiple	$\geq$ aortic valve
Symptoms	Asymptomatic	HF (breathless, faltering growth), recurrent chest infections
Murmur	Loud pansystolic, lower LSE (narrow hole $\rightarrow$ high-speed jet)	Soft pansystolic or none (big hole) + apical mid-diastolic ($\uparrow$ mitral flow)
P2	Quiet	Loud (pulmonary HTN)
Other signs		Tachypnea, tachycardia, cardiomegaly, hepatomegaly, active precordium (visible apex beat)
CXR	Normal	As ASD: cardiomegaly ($> 50\%$), $\uparrow$ pulmonary vascular markings, enlarged PA, LA enlargement (splayed bronchi)

- **Pansystolic** = whole systole; **ejection** = start of systole
- **Loud murmur = small defect**; soft = large

4.3 Small VSD: Management

- **Close spontaneously** (esp. muscular) $\rightarrow$ follow-up echo to check size
- Prevent **endocarditis** (high-speed jet): good **dental hygiene** + prophylactic antibiotic
- Follow the murmur grade:
 - ▶ Grade $\uparrow$ (3 $\rightarrow$ 5) $\rightarrow$ defect smaller $\rightarrow$ **improving**
 - ▶ Grade $\downarrow$ (5 $\rightarrow$ 3) $\rightarrow$ defect larger $\rightarrow$ **not improving**

4.4 Large VSD: Management

1. **Medical**: **diuretics + captopril** for HF; **extra calories** $\rightarrow$ build weight first (FTT $\rightarrow$ can't operate)
 2. **Surgery at 3–6 months** $\rightarrow$ control HF + faltering growth; prevent permanent lung damage
- **Always pulmonary HTN** in large VSD with L $\rightarrow$ R shunt $\rightarrow$ irreversible pulmonary vascular damage

4.5 Eisenmenger Syndrome & Self-Learning Case

- Lungs get high flow $\rightarrow$ pulmonary arteries thicken, stiffen $\rightarrow$ **irreversibly** $\uparrow$ **PVR** $\rightarrow$ right HF
- Reversible early, irreversible later $\rightarrow$ only Rx = **heart-lung transplant** $\rightarrow$ treat large VSD early
- **Case (doctor: a similar question will come)**: 5-wk girl, wheeze, poor feeding + weight gain for 2 wk; RR 50–60, recession, thrill, **pansystolic murmur lower LSE**, accentuated P2, liver 2 cm $\rightarrow$ CXR cardiomegaly + $\uparrow$ markings $\rightarrow$ echo: moderate VSD $\rightarrow$ diuretics + captopril $\rightarrow$ closed at 18 mo
- HF symptoms $\uparrow$ until ~ 3 mo, then may improve as PVR rises $\rightarrow$ untreated $\rightarrow$ **Eisenmenger**

5. Persistent Ductus Arteriosus (PDA) & the Cyanosed Neonate

5.1 PDA: Definition

- Ductus arteriosus: **pulmonary artery** → **descending aorta**; closes shortly after birth (functional, then anatomical)
- PDA = not closed **1 month after expected delivery date** (defective constrictor mechanism)
- After birth PVR falls → flow **aorta** → **PA** (L→R) → SOB
- Blood "stolen" from systemic circulation → ↓ organ flow incl. brain → **risk of stroke**

5.2 PDA: Clinical Features

- **Continuous murmur beneath the left clavicle** = **"machinery" murmur** (most of systole + diastole)
- ↑ Pulse pressure → **collapsing / bounding pulse**
- Symptoms unusual; large duct → ↑ pulmonary flow → HF + pulmonary HTN
- CXR + pathophysiology same as ASD/VSD

5.3 PDA: Management

- **Close it** → removes lifelong risk of **endocarditis** + **pulmonary vascular disease**
- **Coil / occlusion device** via cardiac catheter at **~1 year**
- Occasionally surgical ligation
- **Indomethacin (NSAID)** → ↓ prostaglandins → duct constricts, closes **without opening the chest** (better than surgery)

5.4 Right-to-Left Shunts

- 1) **Tetralogy of Fallot** 2) **Transposition of the great arteries**
- Presentation = **cyanosis**

5.5 Hyperoxia (Nitrogen Washout) Test

- Purpose: cyanosed neonate → **cardiac vs pulmonary** cause?
- ABG on air → then **100% O₂ for 10 min** (headbox/ventilator) → repeat right radial PaO₂
- **PaO₂ stays < 15 kPa (113 mmHg)** → **cyanotic CHD** (once lung disease + PPHN excluded)
- **PaO₂ > 20 kPa** → not cyanotic CHD
- Big rise = pulmonary problem; no change = cardiac problem

5.6 Management of the Cyanosed Neonate (Emergency)

- Same whatever the cause (TOF or TGA)
- 1) **ABC** ± ventilation
- 2) **Prostaglandin infusion 5 ng/kg/min** → keeps ductus open → give **before echo**, buys time
- Most cyanotic CHD in first days = **duct-dependent** → **ductal patency = key to survival**
- PG side effects: **apnea**, jitteriness, seizures, flushing, vasodilatation, hypotension → can't give long-term

6. Right-to-Left Shunts: Tetralogy of Fallot (TOF)

6.1 Anatomy

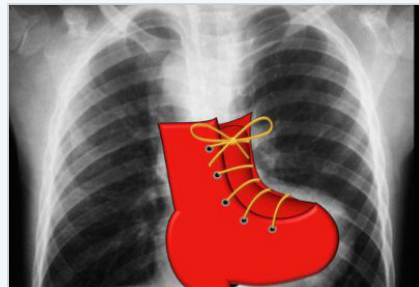
- **Most common cyanotic CHD**
- 4 features:
 1. **Large VSD**
 2. **Overriding aorta** (displaced over the septum)
 3. **Subpulmonary stenosis** → RV outflow obstruction (intermittent → "spells")
 4. **RV hypertrophy** (RV pumps against stenosis)
- Cyanosis is due to the pulmonary stenosis; spectrum from **pink TOF** → **blue TOF**

6.2 Symptoms & Signs

- **Severe cyanosis**
- **Hypercyanotic spells** + **squatting on exercise** (late infancy)
- **Spell** = rapid ↑ cyanosis + irritability / inconsolable crying (hypoxia), breathlessness, pallor (acidosis); **very short murmur during a spell**
- **Emergency** → untreated → MI, stroke, death
- **Squatting** (knee-chest) → ↑ peripheral vascular resistance in legs → more blood RA → RV → PA → relieves cyanosis
- **Clubbing** (older children)
- **Loud harsh ejection systolic murmur, left sternal edge, from day 1**
- ↑ RVOT obstruction → murmur **shortens** + cyanosis **increases**

6.3 CXR: Boot-Shaped Heart

Pulmonary stenosis → ↓ pulmonary flow → **oligemic (dark) lungs**
+ RV pressure overload → RVH → **uptilted apex**
+ concave **pulmonary artery "bay"**
→ **small, boot-shaped heart**



6.4 Management of a Hypercyanotic Spell

- Usually **self-limiting** → followed by sleep; > **15 min** → treat promptly
- First: ABCs → **knee-chest (squatting) position**
- **Sedation + pain relief: morphine** (spells are painful, like MI)
- **IV propranolol** (or α -agonist) → peripheral vasoconstriction + relieves subpulmonary muscle obstruction (same idea as squatting)
- **IV fluids** → ↑ venous return → ↑ pressure into PA
- **Bicarbonate** → correct acidosis (acid-base imbalance can stop the heart)
- **Paralysis + ventilation** → ↓ metabolic O_2 demand

6.5 Surgery

- **Definitive (full corrective) surgery at ~6 months (5–6 mo)**: close VSD + relieve RVOT obstruction (± patch across pulmonary valve)
- Very cyanosed neonate → first a **modified Blalock–Taussig shunt** (subclavian → pulmonary artery) or balloon dilatation of RVOT
- Why? Sudden correction floods **underdeveloped pulmonary arteries** → damage → BT shunt "trains" them = **bridging surgery**
- Order: 1) BT shunt → 2) full correction

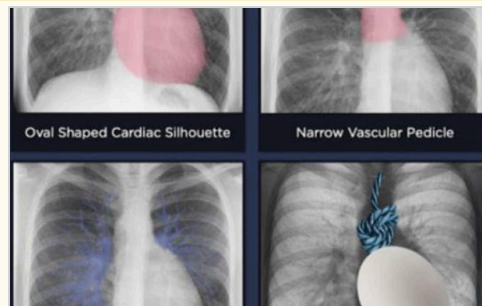
7. Transposition of the Great Arteries (TGA) & Duct-Dependent Neonate

7.1 TGA: Clinical

- **Permanent cyanosis**; cyanosis is the predominant symptom, can be life-threatening
- Presents on **day 2** → ductus closes → ↓ mixing of blue + red blood
- Less severe / later if more mixing (**ASD or VSD** act as the shunt)
- Signs: **cyanosis always** · **loud single S2** · **usually NO murmur** (no shunt) unless ASD/VSD or LV outflow stenosis
- **Scenario**: normal baby discharged → sudden severe cyanosis + collapse (ductus closes) → **give prostaglandin**

7.2 TGA: CXR

- **"Egg on side / egg on a string"**
- Oval-shaped heart
- **Narrow pedicle**
- ↑ **Pulmonary vascular markings**



7.3 TGA: Management

- Sick cyanosed neonate → key = **improve mixing**
- **Prostaglandin to keep the duct open = mandatory**
- **Balloon atrial septostomy** → life-saving in **20%** → not perfect oxygenation but baby lives (75–80% sat); used until transfer if no cardiac surgeon (and no ASD/VSD)
- All need surgery → **arterial switch** in the neonatal period

7.4 Case: Collapsed Neonate (Coarctation)

- 2-day-old, normal check, discharged → **sudden collapse**, pale, grey lips
- **Right brachial pulse weak, femoral pulses impalpable**, big liver, severe metabolic acidosis
- DDx: **CHD** · **sepsis** · **inborn error of metabolism**
- Rx: ventilate + volume · cultures + antibiotics · amino acids / urine organic acids · femorals still absent → **prostaglandin** → pink in 2 h
- Echo: **severe coarctation of the aorta** → **shock from LV outflow obstruction once the duct closed**
- Treated with a **stent** by cardiac catheterization → narrowing abolished

7.5 Key Message

- **Maintaining ductal patency (prostaglandin) is the key to early survival in duct-dependent neonates**
- Cyanotic CHD in the first days (TOF, TGA) → duct-dependent; duct-dependent systemic circulation → shock (coarctation, hypoplastic left heart, critical AS)

Summary

Lesion	%	Key sign	CXR	Management
ASD secundum	7% (80% of ASDs)	ESM upper LSE, fixed wide split S2; often asymptomatic	Cardiomegaly, plethora, big PA	Catheter device (stable → wait to 3–4 y)
ASD primum (partial AVSD)		As above + apical pansystolic murmur	As above	Open surgery at 3–5 y
VSD small	30%	Loud pansystolic lower LSE, quiet P2	Normal	Closes spontaneously; dental hygiene, prophylaxis
VSD large		HF, soft/no murmur, mid-diastolic apical, loud P2	Cardiomegaly, plethora	Diuretics + captopril + calories → surgery 3–6 mo
PDA	12%	Continuous machinery murmur under L clavicle, bounding pulse	As L→R	Indomethacin / coil or device ~1 y / ligation
TOF	5%	Cyanosis, spells, squatting, harsh ESM LSE from day 1, clubbing	Boot-shaped, oligemic	Spells: knee-chest, morphine, propranolol, fluids, bicarb; BT shunt → repair ~6 mo
TGA	5%	Cyanosis day 2, loud single S2, no murmur	Egg on string, narrow pedicle, plethora	PG → septostomy → arterial switch (neonatal)
Coarctation	5%	Collapsed neonate, absent femoral pulses		PG → stent / repair
Complete AVSD	2%	Breathless + blue; Down syndrome		

Key Numbers

Item	Value	Item	Value
CHD incidence	8 / 1000; 30% need Tx in year 1	Innocent murmur	30% of children
Stillborn with cardiac anomaly	1 in 10	Normal O ₂ sat	≥ 94%
Complex / extra-cardiac	10–15% / 10–15%	Cardiothoracic ratio (cardiomegaly)	> 60% (< 1 y), > 50% (> 1 y)
Hyperoxia test	< 15 kPa cardiac; > 20 kPa not cardiac	Prostaglandin	5 ng/kg/min
Spell needing Tx	> 15 min	Surgery timing	VSD 3–6 mo · TOF ~6 mo · ASD 3–5 y · PDA ~1 y · TGA neonatal

Ultra-Quick Differentiation (Keywords)

Keyword combination	Think of
Fixed, widely split S2 + ESM upper LSE	ASD
Wheezing child + abnormal echo / 40 y with arrhythmia	ASD
Loud pansystolic murmur lower LSE, well child	Small VSD
Infant with HF, soft murmur, loud P2	Large VSD
Continuous machinery murmur under left clavicle, bounding pulse	PDA
Sweating + breathless on feeding, big liver	Heart failure
Diastolic murmur	Always pathological
Soft, systolic, left sternal edge, asymptomatic	Innocent murmur
Down syndrome	AVSD
Maternal SLE	Complete heart block
Williams syndrome	Supravalvular aortic stenosis
Turner syndrome	Coarctation, aortic stenosis

Keyword combination	Think of
Most common cyanotic CHD	TOF
Squatting, spells, clubbing	TOF
Boot-shaped heart, oligemic lungs	TOF
Cyanosis on day 2, no murmur, single S2	TGA
Egg on a string, narrow pedicle	TGA
VSD + cyanosis	Think TGA
Collapsed neonate, absent femoral pulses	Coarctation → prostaglandin
PaO ₂ stays < 15 kPa on 100% O ₂	Cyanotic CHD
Untreated large VSD, irreversibly ↑ PVR	Eisenmenger
Apnea after starting a neonatal infusion	Prostaglandin side effect
Close a PDA without surgery	Indomethacin
Very cyanosed neonate with TOF	Blalock–Taussig shunt

Case-Based Practice MCQs (answers on the next page)

1. A 2-day-old, discharged after a normal check, suddenly collapses. He is pale with grey lips, femoral pulses are impalpable and the liver is enlarged. After ventilation and fluids, the most important next step is:
 - A. Indomethacin
 - B. Prostaglandin infusion
 - C. Propranolol
 - D. Oxygen only
2. A 5-week-old has poor feeding, tachypnea 55/min, a pansystolic murmur at the lower left sternal edge, accentuated P2 and a liver 2 cm below the costal margin. Most likely:
 - A. Secundum ASD
 - B. Moderate VSD with heart failure
 - C. TOF
 - D. Innocent murmur
3. A baby was acyanotic and discharged normally, then became severely cyanosed and collapsed. The best immediate treatment is:
 - A. Arterial switch today
 - B. Prostaglandin infusion
 - C. Indomethacin
 - D. Diuretics
4. A child with TOF has a spell lasting 20 minutes. Which is NOT part of management?
 - A. Knee-chest position
 - B. Morphine
 - C. IV propranolol
 - D. Prostaglandin E synthesis inhibitor (indomethacin)
5. A murmur with a fixed, widely split S2 and an ejection systolic murmur at the upper left sternal edge suggests:
 - A. VSD
 - B. ASD
 - C. PDA
 - D. TGA
6. A small VSD murmur changes from grade 5 to grade 3 at follow-up. This means:
 - A. The defect is closing
 - B. The defect is getting larger
 - C. No change
 - D. Endocarditis
7. Which murmur is always pathological?
 - A. Soft systolic
 - B. Diastolic
 - C. Systolic at the left sternal edge
 - D. Ejection murmur in an asymptomatic child
8. A cyanosed neonate's PaO₂ stays at 6 kPa after 10 minutes of 100% oxygen. This suggests:
 - A. Lung disease
 - B. Cyanotic heart disease
 - C. Normal response
 - D. Anemia
9. A continuous machinery murmur beneath the left clavicle with bounding pulses is:
 - A. VSD
 - B. PDA
 - C. Coarctation
 - D. ASD
10. The most common cyanotic congenital heart disease is:
 - A. TGA
 - B. TOF
 - C. AVSD
 - D. Coarctation
11. CXR shows an "egg on a string" heart with a narrow pedicle. Diagnosis?
 - A. TOF
 - B. TGA
 - C. ASD
 - D. Large VSD
12. A baby with Down syndrome most likely has:
 - A. AVSD
 - B. Coarctation
 - C. Supravalvular AS
 - D. Complete heart block
13. A side effect to watch for when starting prostaglandin is:
 - A. Hypertension
 - B. Apnea
 - C. Bradycardia only
 - D. Hyperglycemia
14. A large VSD with heart failure should be operated at:
 - A. Birth
 - B. 3–6 months
 - C. 3–5 years
 - D. Adolescence

Answer Key

1	B: duct-dependent systemic flow (coarctation) → prostaglandin keeps the duct open
2	B: the self-learning case; HF signs + pansystolic murmur lower LSE → VSD
3	B: TGA scenario → PG maintains the ductus; switch comes later
4	D: spell Rx = position, morphine, propranolol, fluids, bicarbonate, paralysis/ventilation
5	B: fixed split S2 = ASD
6	B: lower grade = larger defect = not improving
7	B: diastolic murmurs are always pathological
8	B: $\text{PaO}_2 < 15 \text{ kPa}$ after 100% O_2 → cyanotic CHD
9	B: PDA
10	B: TOF
11	B: TGA
12	A: Down → AVSD (30%)
13	B: apnea (also jitteriness, seizures, flushing, hypotension)
14	B: 3–6 months